Name: Jianyu FU

Position: Ph.D. Student

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Education

Ph.D. in Socio-Cultural Environmental Studies
Graduate School of Frontier Sciences, The University of Tokyo, Japan
2025 – Present

- Research area: Urban perception, mobility data analysis, and AI-based cityscape modeling.
- Supervisor: Prof. Yoshihide Sekimoto
- Member of JST SPRING GX program (Grant No. JPMJSP2108)

M.Arch. in Architecture
Southeast University (Project 985), Nanjing, China
2021.09 – 2024.06

- Grade average: **90.1 / 100 (Rank 1 / 36)**
- Research direction: Urban Architecture, Urban Planning and Design, Urban Analytics, and Big Data Analysis.

B.Arch. in Architecture
Northeastern University (Project 985), Shenyang, China
2015.09 – 2020.06

- Undergraduate thesis focused on sustainable architectural design and human-environment interaction.

Publications & Conference Papers

- **FU, J.**, Wang, X., Ma, J., Zhao, C., Ogawa, Y., & Sekimoto, Y. (2027). *Evaluating urban street views through collective emotional perception from multimodal social media cityscape posts: An R-ECoT-based interpretable framework.*
  Landscape and Urban Planning, 278, 105822. Online 3 Nov 2026. JCR Q1; IF 11.5.
  doi:10.1016/j.landurbplan.2026.105822
- *Beyond Top-Down Urban Sensing: Cross-View Explanatory Asymmetry in Image-Estimated Streetscape Perception in Tokyo.*
  IEEE BigData 2026 (CCF-C), Special Session on Neo Urban Metabolism. Accepted for presentation, 3 Nov 2026 (SP06202).
- Zhang, Y., **FU, J.**, Gao, Y., & Wang, J. (2026). *Comprehensive evaluation method of rooftop views in high-density urban areas: a case study of Guangzhou's historic district.*
  Urban Design International. https://doi.org/10.1057/s41289-025-00287-2
- **FU, J.**, Ma, J., & Sekimoto, Y. (2025). *Urban Real Accessibility Evaluation Based on GPS Data: Integration of Multiple Accessibility Indicators and Deep Learning.*
  Poster accepted for presentation at NetMob 2025, France.
- **FU, J.**, Zhong, Z., Wang, X., Ma, J., & Sekimoto, Y. (2025). *Linking Urban Morphology and Human Activity Patterns: An Empirical Study Based on Mobility Survey Data from the Paris Region.*
  2025 IEEE International Conference on Big Data (BigData), 3685–3694. Oral presentation.
  https://doi.org/10.1109/BigData66926.2025.11402166
- Zhao, F., Xu, X., Dong, W., **FU, J.**, Bai, X., & Wu, J. (2024). *Research on sustainable renewal of historic blocks based on urban catalyst theory.*
  Applied and Computational Engineering, 63(1), 195–205. https://doi.org/10.54254/2755-2721/63/20240974
- **FU, J.** (2022, November). *Research on Urban Public Space Quality Evaluation Model Based on Scene Photo's Deep Learning.*
  the *2022 Computational Design Symposium and Annual Conference of the Computational Design Academic Committee of the Architectural Society of China.*
- **FU, J.** (2022, November). *Spatial Cognitive Preference of Traditional Village Nodes from the Perspective of Eye-Tracking Technology.*
  the *2022 Computational Design Symposium and Annual Conference of the Computational Design Academic Committee of the Architectural Society of China.*

Research Interests

- Urban perception and emotion modeling
- Cityscape analytics and environmental cognition
- AI and deep learning in urban morphology
- Big data analysis for urban design and planning

Technical Skills

- **Programming:** Python (3+ years),
- **Machine Learning & Deep Learning:** PyTorch, TensorFlow, scikit-learn, Keras
- **Spatial Analysis & GIS:** QGIS, ArcGIS, Kepler.gl, PLATEAU 3D City Model
- **Data Processing:** Pandas, GeoPandas, OpenStreetMap & social media APIs
- **Visualization:** matplotlib, seaborn, plotly, Power BI

Professional Experience

Visiting Ph.D. Researcher
Department of Geography, The University of Hong Kong (Host: Prof. Bo Huang)
2026.08.05 – 2026.10.10

- Conducted collaborative research on urban mobility and human behavior, focusing on spatial patterns of individual travel-time burdens in Greater Paris.
- Analyzed the GNSS-based sanitized mobility dataset and urban spatial information to examine heterogeneity in observed travel experiences.
- Research manuscript: “When mobility continues at a cost: Individual travel-time burdens in Greater Paris” (under review).

Sponsored Researcher
SOMAR at ICPSR, University of Michigan
2025.12.15 – 2026.10.15

- Appointed as a sponsored researcher/member of the University of Michigan with access to U-M computing services.
- Granted approved access to Meta Content Library API data through the SOMAR Virtual Data Enclave (VDE) under SOMARAPPLY-768 (SOMAR Data Application).
- Conduct research using publicly available Facebook and Instagram social media data in a secure computing environment for urban perception and collective emotion analysis.

Research Assistant
Sekimoto Laboratory, Center for Spatial Information Science (CSIS), The University of Tokyo
2025 – Present

- Conducted multimodal data analysis on urban perception and mobility behavior.
- Processed large-scale GPS, SNS, and image datasets under strict privacy protocols.
- Participated in SPRING GX interdisciplinary research on sustainable, emotion-aware cities.

Awards & Grants

- JST SPRING GX Fellowship, The University of Tokyo (2025.04–Present)
- NetMob 2025 Poster Track Selection (2025)
- Best Presentation Award at IEEE BigData 2025